

CHM 120

Solutions and Dilutions

Final Report

Student Name	Ryan Saelens
Student ID	235544
Lesson	Solutions and Dilutions
Institution	Rock Valley College
Session	Summer 2022
Course	CHM 120
Instructor	Jason Brinkley

Test Your Knowledge

Match each term with the best definition.

A homogeneous mixture of two or more components.	1	Solution
A solution that contains a greater amount of a dissolved solute than is present in a saturated solution.	2	Supersaturated solution
A solution that holds less than the maximum amount of a solute.	3	Unsaturated solution
A solution that holds the maximum amount of a solute.	4	Saturated solution
The major substance in which a solute is dissolved in a solution.	5	Solvent
The minor substance dissolved in a solution.	6	Solute

Categorize each statement as true or false.

True	False
1 Increasing temperature usually increases the solubility of a solid in a liquid. The solubility of a gas in a liquid decreases with increasing temperature.	2 Decreasing pressure will increase the solubility of a gas in a liquid. The solubility of a solid in a liquid increases with increasing pressure. Increasing pressure will increase the solubility of a solid in a liquid.

Match each concentration with the correct expression.

Amount of solute divided by the amount of solution	1 Concentration
Volume of the solute divided by volume of solution	2 Volume percent
The concentration of the stock solution multiplied by the volume of the stock solution is equal to the concentration of diluted solution multiplied by the volume of the diluted solution	3 Concentration of a Dilution
Mass of the solute divided by the volume of solution	4 Mass/volume percent
Mass of the solute divided by mass of the solution	5 Mass percent

Exploration

When sugar and water are mixed, a clear liquid results. In this scenario, the sugar is considered a ____.

- ☐ solute
- ☐ solution
- ☒ solvent
- ☐ All of the above

The solubility of a gaseous solute in water is ____ by an increase in pressure.

- ☐ increased
- ☒ decreased
- ☐ not affected

When a solution contains less than the maximum amount of dissolvable solute, the solution is said to be unsaturated.

- ☒ True
- ☐ False

The ____ of a solution is the mass of the solute divided by the volume of the solution.

- ☐ volume percent
- ☐ mass percent
- ☐ volume/mass percent
- ☒ mass/volume percent

A dilution is a solution made by adding a solvent to a supersaturated solution.

- ☐ True
- ☒ False

Exercise 1

The normal recipe for preparing Kool-Aid® calls for adding the entire package and 1 cup of sugar to 2 quarts of water. Calculate the volume percent of this solution and determine which of your samples is the closest to the concentration of the recommended preparation. Again, assume that the weight of the drink mix is 0.0 g. The total volume of the solution is 8 and 2/3 cups.

B / U

32 Word(s)

I think cup 2 is the closest to the concentration as the volume percent was measured to be roughly 11.7% and cup 2 has 5.6% (if I did things right that is).

A solute often takes on properties of the solvent. How was this exhibited in Part 1?

B / U

24 Word(s)

The sugar started to mix with the water which really only leaves water and it's properties with the extra mass of the sugar added

List and describe the methods discussed to express the concentration of a solution.

B / U

99 Word(s)

C_1 is the stock solution that was made at the beginning of the lab.
 C_2 is the concentration of each diluted solution that was made (cups 1, 2, 3, and 4) -->
This is what we were solving for!
 V_1 is the volume of the stock solution that differed per cup.
 V_2 is the total volume of the stock solution, at two cups.

I'm not sure if I answered this correctly.

Another answer could be the we discussed Volume% for concentration of solute, and also $C_1V_1 = C_2V_2$ and solved for C_2 to get the concentration of the diluted solution.

Describe the effects of temperature and pressure on solubility. If you had used boiling water instead of lukewarm water, would your results have changed? If so, how?

B / U
25 Word(s)

When the temperature is increased, the solubility usually increases, and an increase in pressure leads to a decrease in solubility, if I undertood things correctly.

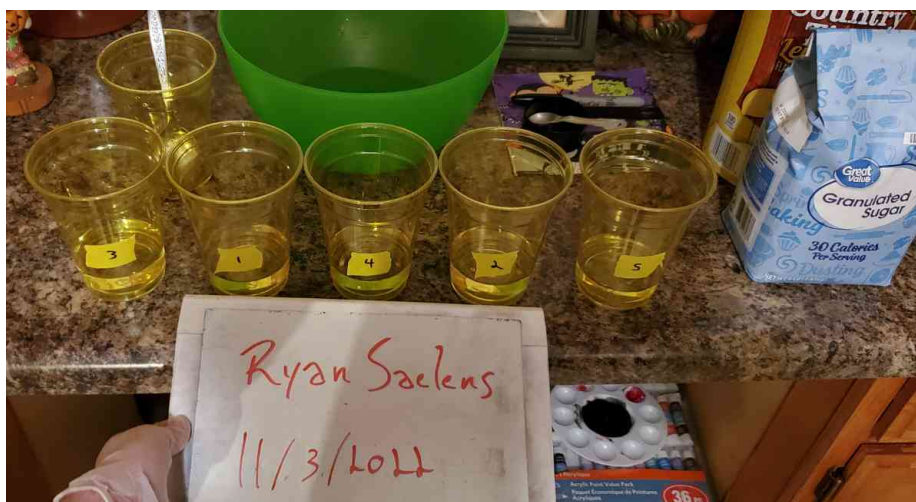
Rubbing alcohol is a commonly used disinfectant and has a cooling effect when applied to the skin. The active ingredient in rubbing alcohol is isopropanol. In drugstores, the most common concentration of rubbing alcohol sold contains 70% (vol/vol) isopropanol in water. Assuming the rubbing alcohol manufacturer uses a 100% isopropanol solution, what volume of pure isopropanol is required to produce a 200-mL bottle of rubbing alcohol? Show all calculations in your answer.

B / U
0 Word(s)

Data Table 1: Concentration and Physical Properties of Drink Mix Solutions

Solution	Concentration (%)	Color Intensity	Taste
Stock Solution (S)	15.1	Clear and somewhat foggy	somewhat sweet but also kind of tasteless
Dilution 1	2.3	clear	very watered down
Dilution 2	5.6	clear; has a little foam around the brim of where liquid meets cup	sweeter, still watered down
Dilution 3	1.9	also clear	by far the smallest taste yet; watered down
Dilution 4	3.7	also clear	somewhat tasteful, still watered down a bit

Photo 1: Drink Mix Color Comparison



Competency Review

The solvent in a solution is the component that ____.

- ☐ gets dissolved
- ☒ dissolves the solute
- ☐ changes phases
- ☐ forms a precipitate

Bronze is a solid solution consisting of 90% copper and 10% tin. Tin is considered the solute of this solution.

- ☒ True
- ☐ False

The solubility of salt is 35.7 g per 100 g of water at 25.0°C. To find the percentage of salt in a saturated solution, which concentration calculation should be used?

- ☐ volume mass percent
- ☐ mass percent
- ☒ volume percent
- ☐ mass volume percent

An increase in temperature ____ the solubility of a solid solute dissolving in a liquid solvent.

- ☒ increases
- ☐ decreases
- ☐ does not influence

A homogeneous mixture of two or more substances is called a(n) ____.

- ☐ colloid
- ☐ compound
- ☐ aggregation
- ☒ solution

A ____ solution contains the maximum amount of dissolved solute under stable conditions.

- ☐ concentrated
- ☐ supersaturated
- ☐ unsaturated
- ☒ saturated

An increase in ____ will increase the solubility of a gas in water.

- ☐ temperature
 - ☒ pressure
 - ☐ water volume
 - ☐ gas volume
-

A 60% alcohol solution is composed of ____ mL of alcohol and ____ mL of water.

- ☐ 60; 100
- ☒ 60; 40
- ☐ 40; 100
- ☐ 40; 60

To dilute a stock solution, the solution and additional solvent are combined.

- ☒ True
- ☐ False

____ solutions contain more solute than could be dissolved under normal conditions.

- ☐ Saturated
- ☐ Unsaturated
- ☐ Aqueous
- ☒ Supersaturated

Extension Questions

In Denver, Colorado the elevation is about 5,280 feet above sea level. Explain what potential effects this may have on the solubility of a gaseous solute in a liquid solution.

Due to it being at the elevation that it is, the atmosphere at the height may have something to do with it. The temperature at that level will play a role in effecting the solubility of a gaseous solute. The liquid may also start to evaporate quicker, that is just a side thought...